PREP-RASHAN

A Unit of Prep-Rashan Tutorial

DAILY CURRENT AFFAIRS - RASHAN

Comprehensive Compilation for UPSC Civil Services Examination

7 November 2026

General Studies Paper I • Paper II • Paper III • Paper IV

Keywords Defined • PYQ (Prelims & Mains) • Practice Questions (Prelims & Mains)

TABLE OF CONTENTS

GS-I: Paper I 3
1. Marine Heatwaves Across India's Seas: When the Ocean Runs a Fever 3
2. Update: The Manuscripts Behind India's New UNESCO Memory Entries 5
3. Virunga National Park: Anthrax Strikes Africa's Oldest Park 7

GS-II: Paper II 9
4. The 24-Hour Clock Starts When Liberty Ends, Not When Arrest Is Written 9
5. Kochon Prize 2026: Indian Non-Profit REACH Honoured for TB Work 11
6. Update: Abdul El-Sayed Reported as the First Muslim US Senator 13
7. France's National Order of Merit for Dr Devi Shetty 15

GS-III: Paper III 17
8. Delhi's Winter Smog: Why GRAP Alone Cannot Clear the Air 17
9. CERT-In's First Audit Rulebook for AI Agents 19
10. Pralay: India's Quasi-Ballistic Battlefield Missile Tested Again 21
11. Teghaza 001: A Pink Rock From Mars's Earliest Crust 23
12. Global Innovation Index 2026: India at 38th, Top of Its Income Group 25
13. Exercise VAJRA-SAT: Rehearsing for a Foot-and-Mouth Strain India Has Never Seen 27
14. RAHAT MITRA 2026: India and Myanmar's First Tri-Services Disaster Drill 29
15. Update: Amini Joins the Arnala Class of Submarine Hunters 31
16. Update: Tokamak – Origins, Heating and Safety 33

GS-IV: Paper IV 35
17. Ethics Synthesis: Liberty, Care and Responsible Machines 35

GENERAL STUDIES PAPER I

[GS-I] Indian Heritage & Culture, History, Geography of the World and Society

GS-I | Geography – Oceanography and Climate

1. Marine Heatwaves Across India's Seas: When the Ocean Runs a Fever

Syllabus: *GS-I: Important geophysical phenomena; distribution of key natural resources. GS-III: Climate change; disaster management.*

CONTEXT

INCOIS, Hyderabad warned of a **marine heatwave** across India's seas – the **Bay of Bengal**, the **Arabian Sea** and the wider **Indian Ocean** – with surface waters running up to about **1°C** warmer than usual, along with **swell surge** alerts for coastal States.

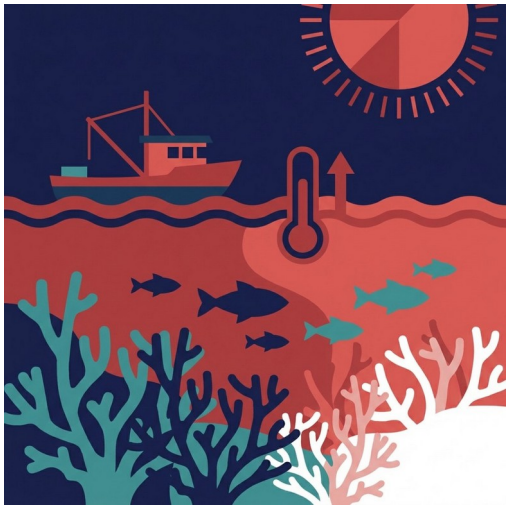


Illustration: a coral reef turning white under a hot sun, a rising thermometer over red sea water, fish moving away and a small fishing boat.

KEYWORDS DEFINED

Keyword	Meaning
Marine heatwave (MHW)	A spell of **unusually warm sea surface temperature** – above the **90th percentile** of normal for **at least five days**.
INCOIS	**Indian National Centre for Ocean Information Services**, Hyderabad – under the **Ministry of Earth Sciences**; gives ocean forecasts and alerts.
Mixed layer	The **well-stirred top layer** of the ocean; a **shallow** one heats up **fast**.
Zooxanthellae	**Algae living inside corals** that give them food and colour; heat makes corals **expel** them (**bleaching**).
Indian Ocean Dipole (IOD)	A **see-saw of sea temperatures** between the **western and eastern** Indian Ocean.

What a marine heatwave is

- Sea temperature stays **above the 90th percentile** of the local normal for **five days or more**.
- INCOIS's **MAHAS** system tracks heatwaves over **6 ocean basins** and **14 Indian coastal sectors**, using **Watch** and **Alert** levels.
- The **tropical Indian Ocean** is warming fast; heatwaves in the **western Indian Ocean** have risen **about fourfold** in recent decades.
- **Swell waves above 1 metre** were forecast for **11 coastal States and UTs**.

How they form

- **Strong sunshine and clear skies** under **high-pressure ridges** heat the surface.
- **Weak winds** stop warm surface water mixing with **cooler deep water**.
- A **shallow mixed layer** traps heat near the top.
- **Currents** carry warm water into new areas; **El Niño** or a **positive IOD** add to the warming, on top of **climate change**.

What they do

- **Coral bleaching** – reefs of the **Gulf of Mannar, Palk Bay, Lakshadweep and Andamans** at risk.
- **Fish move** deeper or away – **sardine and mackerel** catches fall off **Kerala and Karnataka**.
- **Harmful algal blooms ('red tides')** use up oxygen; **seagrass** dies back, hurting **dugongs and turtles**.
- Extra ocean heat can **rapidly strengthen cyclones** over the Arabian Sea and Bay of Bengal.

Way forward

- More **ocean observation** (buoys, Argo floats, satellites) and **early warnings** built into **disaster plans**.
- **Sustainable fishing**, reef protection and **alternative livelihoods** for fishers in bad seasons.
- **Value addition:** India's **Deep Ocean Mission** and the **Ocean Services, Modelling, Application, Resources and Technology (O-SMART)** scheme fund ocean research *(general knowledge)*.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **coral bleaching, ENSO/IOD** and **ocean-monitoring bodies** such as INCOIS.

PYQ – MAINS

- **Pattern:** Mains asks how **ocean warming** affects **cyclones, fisheries and coastal livelihoods**.

FQ – PRELIMS (Possible Question)

- Consider: 1. A marine heatwave lasts at least five days above the 90th-percentile threshold. 2. INCOIS works under the Ministry of Earth Sciences. 3. A deep mixed layer makes surface waters heat up faster. Which are correct? **(Ans: 1 and 2 only)**

FQ – MAINS (Possible Question)

- "Marine heatwaves are the silent disasters of a warming ocean." Discuss their causes and impacts on India. (150 words)

GS-I | Art & Culture – Documentary Heritage UPDATE

2. Update: The Manuscripts Behind India's New UNESCO Memory Entries

Syllabus: *GS-I: Indian culture – literature and heritage. GS-II: International institutions (UNESCO).*

WHAT WE KNOW TILL NOW

- **Kalhana's Rajatarangini**, the **Mahabharata** and **Kautilya's Arthashastra** joined UNESCO's **Asia-Pacific Memory of the World register** at the **11th MOWCAP** meeting in **Jakarta**.
- UNESCO's **Memory of the World** programme began in **1992** and keeps **international, regional and national** registers.

WHAT TODAY ADDS

- **The physical manuscripts:** the Rajatarangini copy is in **Sharada script on birch bark**; the Arthashastra copy is a **16th-century palm-leaf manuscript in Grantha script** at the **Oriental Research Institute, Mysuru**.
- India's **earlier documentary entries**, including the **Bhagavad Gita and Natyashastra (2025)**.

CONTEXT

More details have emerged on the three Indian texts placed on UNESCO's **Asia-Pacific Memory of the World** register – the **physical manuscripts** that were nominated and India's **earlier entries**.



Illustration: an old manuscript and a bound book on a stone pedestal under a glowing lamp – heritage kept safe for the future.

KEYWORDS DEFINED

Keyword	Meaning
Sharada script	An old **Brahmi-derived script of Kashmir**, used for Sanskrit and Kashmiri.
Grantha script	A **South Indian script** used to write **Sanskrit**, especially in Tamil regions.
Bhurja-patra	**Birch bark** – the traditional writing surface of **Kashmir and the Himalaya**.
Oriental Research Institute,	A manuscript library (**1891**) that holds the **Arthashastra** copy rediscovered

Keyword	Meaning
Mysuru	by **R. Shamasastry**.
Kavya	**Sanskrit poetic literature** – the Rajatarangini is history written as poetry.

Rajatarangini – the nominated text

- Written by **Kalhana**, a **12th-century Kashmiri** historian, around **1148–1149 CE**.
- Copy in **Sharada script** on **birch bark**; about **7,826 verses** in **eight Tarangas**.
- History told in the form of **kavya** – kings, dynasties, politics and society of **Kashmir**.

Arthashastra – the nominated text

- A **Sanskrit treatise** on **statecraft, economy, law, spies and administration**, linked to **Kautilya** of the **Mauryan court** (**4th century BCE** by tradition).
- The nominated copy is a **complete palm-leaf manuscript of the 16th century** in **Grantha script**.
- Preserved at the **Oriental Research Institute, Mysuru**, where **R. Shamasastry** found it.

Mahabharata – the nominated text

- Attributed to **Vyasa**; grew from **Jaya** to **Bharata** to the full epic.
- Explores **duty, justice, power and kinship**; the **Bhagavad Gita** lies within it.

India on the Memory of the World map

- Earlier Indian entries include the **Rigveda manuscripts**, the **Dutch East India Company archives**, the **works of Abhinavagupta** and the **archives of the first NAM summit (Belgrade, 1961)**.
- The **Bhagavad Gita** and **Bharata's Natyashastra** were added to the **international register in 2025**.
- **Value addition:** the **National Mission for Manuscripts (2003)** documents India's manuscripts; the **Gyan Bharatam Mission** aims to digitise them *(general knowledge)*.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **scripts, texts and where manuscripts are kept**, and UNESCO lists.

PYQ – MAINS

- **Pattern:** Mains asks about **preserving and digitising** India's manuscript heritage.

FQ – PRELIMS (Possible Question)

- The palm-leaf manuscript of the Arthashastra preserved at the Oriental Research Institute, Mysuru, is written in which script? (a) Sharada (b) Grantha (c) Kharoshthi (d) Modi **(Ans: b)**

FQ – MAINS (Possible Question)

- "Manuscripts are India's memory; digitisation is its insurance." Discuss. (150 words)

GS-I | Geography – Places in News (Mapping); Zoonoses

3. Virunga National Park: Anthrax Strikes Africa's Oldest Park

Syllabus: *GS-I: Important geographical features (mapping). GS-III: Conservation; zoonotic diseases.*

CONTEXT

Anthrax has killed **30 or more hippos** in Congo's **Virunga National Park** since September, prompting **tighter surveillance** to stop spread to people.



Illustration: a rift-valley park with a smoking volcano, hippos in a lake, a gorilla in the forest and a ranger with a sample kit.

KEYWORDS DEFINED

Keyword	Meaning
Albertine Rift	The **western branch** of Africa's **Great Rift Valley**, rich in lakes, volcanoes and species.
Anthrax	A **zoonotic disease** caused by the bacterium *__Bacillus anthracis__, *whose spores** survive in soil for decades.
Zoonosis	A disease that **jumps from animals to humans**.
Eschar	The **painless black-centred sore** typical of **skin anthrax**.
ICCN	Congo's **Institute for the Conservation of Nature**, which co-manages Virunga with the **Virunga Foundation**.

Where Virunga is (mapping)

- In **eastern DR Congo**, along the **Albertine Rift**, next to **Uganda and Rwanda**; about **7,800 sq km**.
- **Africa's oldest national park** – set up in **1925** as **Albert National Park**; renamed **Virunga in 1969**.
- **UNESCO World Heritage Site since 1979**.

What makes it special

- Habitats from **savanna and wetland** to **forest and glacier-capped Rwenzori Mountains**.
- Two of Africa's most active volcanoes – **Nyiragongo and Nyamuragira**.
- Home to **mountain gorillas**, **eastern lowland gorillas**, **chimpanzees**, **okapis**, forest elephants and **hippos**; includes **Lake Edward**.

About anthrax

- Caused by *__Bacillus anthracis__, *a rod-shaped bacterium that forms tough spores** in soil.
- **Droughts and heavy rain** expose spores to **grazing animals**.
- **Three forms: skin (cutaneous)**, **gut (gastrointestinal)** from bad meat, and **inhalation** – the deadliest.
- It **rarely spreads person to person**; treated with antibiotics such as **ciprofloxacin or doxycycline**.

Why it matters

- Shows the **One Health** link – **wildlife, livestock and people** share disease risks.
- **Map tip: Lake Victoria** lies east of Uganda; **Lake Tanganyika** south of the Albertine Rift; **Rwanda's Volcanoes NP** adjoins Virunga *(general knowledge)*.
- **Value addition:** India reports anthrax mostly in **Odisha, Andhra Pradesh and Kerala**, often linked to **handling dead cattle** *(general knowledge)*.

PYQ – PRELIMS

- **Pattern:** Prelims mapping asks about **African parks, lakes and rift valleys**, and about **zoonotic diseases**.

PYQ – MAINS

- **Pattern:** Mains asks about the **One Health** approach to zoonoses.

FQ – PRELIMS (Possible Question)

- Virunga National Park, recently in news for an anthrax outbreak, borders which two countries? (a) Kenya and Tanzania (b) Uganda and Rwanda (c) Burundi and Zambia (d) Sudan and Ethiopia **(Ans: b)**

FQ – MAINS (Possible Question)

- "Zoonoses demand a One Health response." Discuss with an example. (150 words)

GENERAL STUDIES PAPER II

[GS-II] Governance, Constitution, Polity, Social Justice & International Relations

GS-II | Polity – Fundamental Rights (Articles 21 and 22); Criminal Justice

4. The 24-Hour Clock Starts When Liberty Ends, Not When Arrest Is Written

Syllabus: *GS-II: Fundamental rights; judiciary; criminal justice reforms.*

CONTEXT

The **Supreme Court** ruled that the **24-hour limit under Article 22(2)** for producing a person before a **magistrate** starts **the moment liberty is actually restricted**, not when **formal arrest** is recorded – in a case under the **NDPS Act** from **Tripura**.

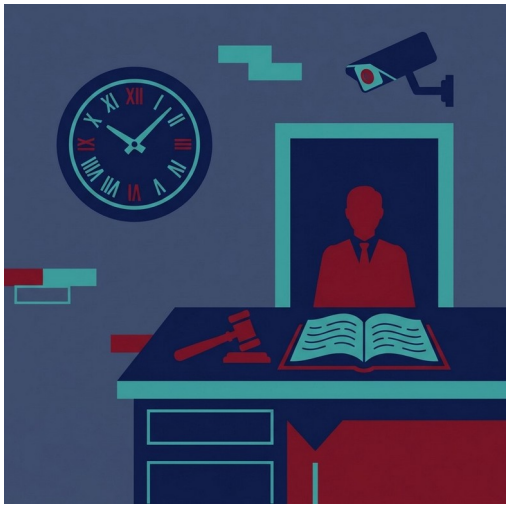


Illustration: a 24-hour clock beside a magistrate's desk with a gavel and the Constitution, a person behind a door and a CCTV camera above.

KEYWORDS DEFINED

Keyword	Meaning
Article 22(2)	Every person **arrested and detained** must be produced before the **nearest magistrate within 24 hours** (travel time excluded).
Detention	Any **real restriction on a person's liberty**, whether or not called 'arrest'.
Remand	A magistrate's order allowing **further custody** – police or judicial.
NDPS Act, 1985	The law on **narcotic drugs**; **Section 37** makes **bail harder** for serious offences.
BNSS, 2023	**Bharatiya Nagarik Suraksha Sanhita** – the new criminal procedure code; **Section 58** bars holding a person **longer than reasonable**.

The case

- **Customs officers in Tripura** stopped people carrying about **60 kg of methamphetamine tablets**.
- They were under the team's **physical control**, but **formal arrest** was recorded only **the next day**.

- The agency said the clock starts at **formal arrest**; the accused said it starts at **first loss of liberty**.
- The Court **set aside** the High Court's **refusal of bail**.

What the Court held

- The **24 hours** run from when liberty is **effectively curtailed** – magistrates must **record that time**.
- **'Arrested and detained'** in Article 22(2) are read **together**, as the same thing for the deadline.
- **No investigative pause** – search, seizure or questioning **cannot stop the clock**; it cited **Justice V.R. Krishna Iyer** in **Niranjan Singh (1980)**.
- Views of the **Orissa, Kerala and Madras High Courts** allowing pre-arrest time to be excluded were **disapproved**.
- Anyone held **beyond 24 hours** without production must be **released at once**.

Further safeguards

- Meeting the deadline **does not make custody reasonable** – magistrates must check **grounds and duration**.
- An **unexplained 18-hour gap** between search and arrest can justify **refusing police remand**.
- **Article 21 prevails** over **Section 37 NDPS** limits where liberty is violated – citing **K.A. Najeeb (2021)**.
- **CCTV with audio, video and night vision** in police stations and offices of **NIA, CBI, ED**; footage kept **at least 18 months**.

Significance

- A **clear start point** for the deadline, whatever label agencies use.
- Stronger **judicial oversight** of **remand** and a check on **custodial violence**.
- **Value addition: D.K. Basu v. State of West Bengal (1997)** laid down arrest guidelines; **Paramvir Singh Saini (2020)** ordered CCTV in police stations and central agencies *(general knowledge)*.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **Articles 20–22**, **preventive detention** and landmark **arrest-safeguard cases**.

PYQ – MAINS

- **Pattern:** Mains asks about **custodial violence**, **police reforms** and **personal liberty**.

FQ – PRELIMS (Possible Question)

- Consider: 1. The 24-hour period under Article 22(2) begins with formal recording of arrest. 2. Time needed to travel to the magistrate is excluded from the 24 hours. 3. Section 37 of the NDPS Act extends the period of police custody. Which is/are correct? **(Ans: 2 only)**

FQ – MAINS (Possible Question)

- "The label of arrest cannot decide when constitutional safeguards begin." Examine the recent ruling on Article 22(2). (250 words)

GS-II | Social Sector – Health (Tuberculosis); Awards

5. Kochon Prize 2026: Indian Non-Profit REACH Honoured for TB Work

Syllabus: *GS-II: Issues relating to health; role of civil society and NGOs.*

> **CONTEXT**
>
> **REACH**, a Chennai-based non-profit, was **jointly awarded the Kochon Prize 2026** for its work against **tuberculosis (TB)** – the **first Indian civil-society organisation** to win it.



Illustration: a medal and trophy beside a lung symbol, a microscope, tablets and a community health worker with a clipboard.

KEYWORDS DEFINED

Keyword	Meaning
Kochon Prize	An **annual global award (since 2006)** for work against **TB**, given by Korea's **Kochon Foundation** with the **Stop TB Partnership**.
Stop TB Partnership	A **UN-hosted global partnership** working to **end TB**.
REACH	**Resource Group for Education and Advocacy for Community Health** – works on **people-centred TB care** in India.
Ni-kshay Poshan Yojana	**Monthly nutrition support** for TB patients through **direct benefit transfer** *(general knowledge)*.
MDR-TB	**Multi-drug-resistant TB** – not cured by the two main first-line drugs.

About the prize

- Given by the **Kochon Foundation**, an independent **South Korean** non-profit, with the **Stop TB Partnership**.
- **Annual**, set up in **2006** in memory of **Chairman Chong-Kun Lee**.
- Carries **US$ 65,000** – the **largest award dedicated only to TB**.

The 2026 award

- Presented in **Indonesia** on **5 November 2026**.
- Theme: **integrating TB care into other health services** for **early diagnosis and treatment**.

- **Joint winners: REACH** from India, Kenya's **Centre for Health Solutions**, and the **TB programme of the Marshall Islands**.

About REACH

- Working for **over 27 years** on **integrated, people-centred TB care**.
- **First Indian civil-society non-profit** to receive the Kochon Prize.
- Focus on **community support, patient advocacy** and linking care across services.

India's TB fight

- India carries the **largest share of the world's TB cases**.
- Government programmes: **National TB Elimination Programme**, **Ni-kshay** portal, **Ni-kshay Mitra** community support.
- **Value addition:** India aims to **end TB by 2025–26** under the **National Strategic Plan**, ahead of the **SDG 2030** target; **TB Mukt Bharat** is the campaign *(general knowledge)*.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **health awards, global partnerships** and **disease programmes**.

PYQ – MAINS

- **Pattern:** Mains asks about **community participation** and **NGOs** in public health.

FQ – PRELIMS (Possible Question)

- The Kochon Prize, recently won by an Indian non-profit, recognises contributions to fighting: (a) malaria (b) HIV/AIDS (c) tuberculosis (d) polio **(Ans: c)**

FQ – MAINS (Possible Question)

- "Ending TB needs communities, not just clinics." Discuss the role of civil society. (150 words)

GS-II | International Relations – US Politics; Indian Diaspora UPDATE

6. Update: Abdul El-Sayed Reported as the First Muslim US Senator

Syllabus: *GS-II: Effect of policies and politics of other countries on India; Indian diaspora.*

WHAT WE KNOW TILL NOW

- The **2026 US midterms** decided **all 435 House seats** and about **one-third of the Senate**; the **Democrats** won the **House** and the **Senate**.
- US **senators** serve **six-year terms**; each State elects **two**.

WHAT TODAY ADDS

- Democrat **Abdul El-Sayed** is reported to have won **Michigan's Senate seat**, becoming the **first Muslim** elected to the **US Senate**.
- His **wife's family** traces its roots to **Bhatkal, Karnataka**.

CONTEXT

Reports say **Abdul El-Sayed**, a **Democrat** and **physician**, won **Michigan's Senate race** in the **2026 midterms**, defeating **Republican Mike Rogers** – the **first Muslim** to be elected to the **US Senate**.

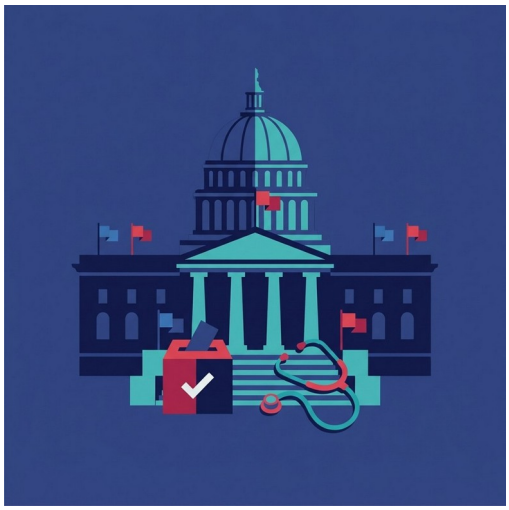


Illustration: a domed legislature building with a ballot box and tick mark, small flags and a stethoscope on the steps.

KEYWORDS DEFINED

Keyword	Meaning
US Senate	The **upper house** of Congress – **100 members**, **two per State**, **six-year terms**.
Midterm elections	US polls held **halfway through a presidential term**.
Indian diaspora	People of **Indian origin or citizenship** living abroad – about **3.5 crore** worldwide *(general knowledge)*.
Bhatkal	A coastal town in **Uttara Kannada district, Karnataka**.
Public health	The science of **protecting the health of whole populations**.

Who he is

- An **American physician and public-health expert** of **Egyptian descent**.
- Studied at the **University of Michigan** and **Oxford**; **medical degree from Columbia University**.
- Former **Health Director of Detroit**, known for work on **health access** in poor communities.

The India link

- His wife, **Sarah Jukaku**, a **psychiatrist**, has family roots in **Bhatkal, Uttara Kannada, Karnataka**.
- Her grandfather **J.H. Shamsuddin Jukaku** was a **Mysore State legislator** and **Deputy Minister** for **Finance and Electricity**.

Why it matters

- A milestone for **minority representation** in US politics.
- Shows growing participation of **immigrant-background** communities in **US governance**.
- Highlights **people-to-people links** between India and the US.

Exam angle

- **GS-II:** India–US ties, **diaspora diplomacy**.
- **GS-IV: diversity, inclusion** and **public-service leadership**.
- **Value addition:** Indian-Americans in Congress have included **Ami Bera, Ro Khanna, Pramila Jayapal and Raja Krishnamoorthi**; **Pravasi Bharatiya Divas** is held on **9 January** *(general knowledge)*.

PYQ – PRELIMS

- **Pattern:** Prelims asks about the **US political system** and **diaspora initiatives**.

PYQ – MAINS

- **Pattern:** Mains asks about the **role of the Indian diaspora** in foreign relations.

FQ – PRELIMS (Possible Question)

- Bhatkal, mentioned in news in connection with a US Senator's family, is located in: (a) Kerala (b) Goa (c) Karnataka (d) Maharashtra **(Ans: c)**

FQ – MAINS (Possible Question)

- "The diaspora is a bridge of soft power." Discuss with examples. (150 words)

GS-II | Social Sector – Health; Awards (India–France)

7. France's National Order of Merit for Dr Devi Shetty

Syllabus: *GS-II: Health; India's relations with other countries (France). GS-IV: Public service and compassion.*

CONTEXT

Dr Devi Prasad Shetty, who founded **Narayana Health**, received France's **National Order of Merit** at the rank of **Chevalier (Knight)** in **Bengaluru**.



Illustration: a heart with an ECG line inside a hospital outline, a surgical light, and a blue six-armed star medal on a ribbon.

KEYWORDS DEFINED

Keyword	Meaning
Ordre national du Mérite	**France's National Order of Merit** – its **second-highest** national honour after the **Legion of Honour**.
Chevalier	**Knight** – the **first rank** of the order.
Economies of scale	**Lower cost per unit** as **volume rises** – used to cut surgery costs.
Yeshasvini scheme	Karnataka's **cooperative health insurance for farmers**, started in **2003**.
Tertiary care	**Specialised hospital care** such as heart surgery.

About the honour

- Created by **President Charles de Gaulle** on **3 December 1963**, replacing **16 ministerial orders**.
- **Five classes:** ranks **Chevalier, Officier, Commandeur**; dignities **Grand Officier, Grand-Croix**.
- Symbol: a **six-armed blue-enamelled star** with **oak leaves** on a **blue ribbon**.
- Given to **French citizens and distinguished foreigners**; has **gender-parity** rules.

About Dr Shetty

- **Cardiac surgeon** and founder of **Narayana Health** (formerly **Narayana Hrudayalaya**), Bengaluru.
- Made heart surgery **affordable** through **high volumes** – called the **'Henry Ford of heart surgery'**.
- Performed India's **first neonatal open-heart surgery (1992)** on a **21-day-old** baby.

Health innovation

- Pioneered Karnataka's **Yeshasvini** micro-insurance for farmers – low premiums, **cooperative** model.
- Shows how **process innovation** can cut costs of **tertiary care**.

Honours and links

- **Padma Shri (2004)**, **Padma Bhushan (2012)**, **Dr B.C. Roy Award**, **Nikkei Asia Prize**.
- Reflects **India–France** ties in **health and science**.
- **Value addition:** the **Legion of Honour (1802, Napoleon)** is France's highest honour *(general knowledge)*.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **foreign honours** and **Indian health schemes**.

PYQ – MAINS

- **Pattern:** Mains asks how **innovation and scale** can make healthcare affordable.

FQ – PRELIMS (Possible Question)

- The National Order of Merit, recently conferred on an Indian surgeon, is an honour of: (a) Germany (b) the United Kingdom (c) France (d) Italy **(Ans: c)**

FQ – MAINS (Possible Question)

- "Affordable quality care needs both innovation and scale." Discuss with examples. (150 words)

GENERAL STUDIES PAPER III

[GS-III] Technology, Economic Development, Bio-diversity, Environment, Security & Disaster Management

GS-III | Environment – Air Pollution; Governance

8. Delhi's Winter Smog: Why GRAP Alone Cannot Clear the Air

Syllabus: *GS-III: Environmental pollution and degradation. GS-II: Governance; inter-State coordination.*

CONTEXT

Delhi's air slipped into the **'Very Poor'** band despite the **Winter Action Plan** (from **1 November**) with curbs on **non-compliant vehicles**, **'No PUCC, No Fuel'** and **construction** – raising the case for **year-round, airshed-wide** action beyond **GRAP**.



Illustration: a hazy skyline under grey smog, crop-burning smoke in distant fields, traffic, a red air-quality warning light and a satellite overhead.

KEYWORDS DEFINED

Keyword	Meaning
GRAP	**Graded Response Action Plan** – **emergency curbs** that tighten as the **AQI worsens**; run by the **CAQM**.
Airshed	The **whole area** whose emissions **mix and affect** a city's air – crossing administrative borders.
CAQM	**Commission for Air Quality Management** in NCR and adjoining areas (statutory, **2021 Act**).
PM2.5	Fine particles **2.5 micrometres or smaller** that reach **deep into the lungs**.
Geostationary satellite	Stays over **one spot** and watches it **all day** – catches fires that polar satellites miss.

Why winter air gets worse

- **Slow winds** let pollutants **pile up**; a **milder, delayed winter** (linked to a strong **El Niño**) shifted the peak season by about **two weeks**.

- Fewer **western disturbances** may mean **less rain** to wash the air.
- **Local sources: traffic jams**, bad roads, **open waste burning** and **construction dust**.

The stubble-burning data gap

- Fires in **Punjab and Haryana** shifted **later in the day – over 90% after 3 PM** in **2024 and 2025**, against **under 3% in 2021**.
- NASA's **MODIS** passes miss these late fires, so official counts are **too low**.
- Korea's **geostationary GK-2A satellite** shows **far more fires**; with realistic data, stubble gave about **20% of Delhi's PM2.5** on **1 November**.

Limits of GRAP

- Like an **ICU**: it **stabilises** a crisis but **does not cure** the causes.
- Repeated curbs cause **public hardship** without **lasting cuts** in emissions.
- Delhi-only action covers just **50–70%** of the sources affecting the city.

Way forward

- **Airshed-based planning** across Delhi, Haryana, Punjab and UP – an indigenous framework, **AIRSHEDEM**, aims at **near-full source coverage**.
- Use **geostationary satellites** and **updated ground data** for emission inventories and forecasts.
- Year-round fixes: **traffic management, road repair, waste and dust control**, crop-residue solutions.
- **Value addition:** the **National Clean Air Programme (2019)** targets **40% lower particulate levels by 2025–26** in **131 cities**; the **AQI** has **six categories** *(general knowledge)*.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **GRAP, CAQM, NCAP, AQI** and **pollutants**.

PYQ – MAINS

- **Pattern:** Mains asks why **air pollution** persists and how to **coordinate across States**.

FQ – PRELIMS (Possible Question)

- Consider: 1. GRAP addresses long-term structural causes of pollution. 2. Geostationary satellites can observe a region continuously through the day. 3. Most stubble fires in Punjab in 2024–25 reportedly occurred after 3 PM. Which are correct? **(Ans: 2 and 3 only)**

FQ – MAINS (Possible Question)

- "Delhi needs an airshed approach, not just emergency curbs." Discuss. (250 words)

GS-III | Internal Security – Cyber Security; Science & Technology (AI)

9. CERT-In's First Audit Rulebook for AI Agents

Syllabus: *GS-III: Challenges to internal security through communication networks; basics of cyber security; awareness in IT and AI.*

CONTEXT

CERT-In issued the **AI Security Auditing Guidelines** – a framework for **secure, ethical and auditable AI** – setting audit baselines especially for **agentic AI** that can plan and act on its own.



Illustration: a robot agent linked to many tools and windows, a padlock shield, an auditor's magnifying glass and a large red emergency-stop button.

KEYWORDS DEFINED

Keyword	Meaning
CERT-In	**Indian Computer Emergency Response Team** – the national agency for **cyber incidents**, under **MeitY**, set up under **Section 70B of the IT Act**.
Agentic AI	AI that can **break a goal into steps**, **use tools and APIs**, run code and **hand tasks to sub-agents**.
Prompt injection	**Hidden malicious instructions** in inputs that **hijack** an AI system.
Human-in-the-loop (HITL)	A **person must approve** important AI actions.
Least privilege	Give a system **only the access it needs** – nothing more.

Agentic vs conventional AI

- **Conventional AI:** answers a prompt – summarising, translating, Q&A – **without acting** on its own.
- **Agentic AI: plans**, calls **external tools**, writes and runs **code**, and **delegates** to other agents.
- More autonomy means **bigger security risks**.

Core of the framework

- **Four risk tiers** – **Low, Moderate, High, Critical** – by autonomy, impact, data sensitivity and recoverability.
- The tier sets **audit depth**, **who audits** (internal or **CERT-In-empanelled** auditors) and how much **human oversight**.

- **Nine audit areas** – governance, data and privacy, model risk, **bias and fairness**, **explainability**, **security and incident reporting**, monitoring, **human oversight**, and **third-party models and APIs**.
- **Seven principles** aligned with **MeitY's AI governance** approach – **security-by-design**, accountability, transparency, fairness, privacy, **auditability**, resilience.

Special rules for AI agents

- **Least privilege** – no unapproved tools or permission escalation.
- Protect **API keys and credentials** from leaking.
- Govern **persistent memory** to stop **memory poisoning** and data leaks.
- **Authenticate sub-agents** so injected prompts cannot spread.
- A **tamper-proof 'kill switch'** to stop any autonomous agent instantly.

Challenges and way ahead

- Agents behave **non-deterministically**, so audits are hard to **repeat**.
- **Indirect prompt injection** via websites, emails and documents.
- Fault-tracing across **multi-agent chains**; **compliance costs** for startups.
- **Way ahead: sandboxed agents**, **human approval gates** for high-impact actions, national **red-teaming testbeds**, **audit vouchers** via the **IndiaAI Mission**.
- **Value addition: ISO/IEC 42001 (2023)** is the global **AI management system** standard; CERT-In requires cyber incidents to be **reported within 6 hours** *(general knowledge)*.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **CERT-In**, the **IT Act**, cyber terms and **AI governance**.

PYQ – MAINS

- **Pattern:** Mains asks about **regulating AI** and **cyber security** frameworks.

FQ – PRELIMS (Possible Question)

- Consider: 1. CERT-In's AI guidelines classify AI systems into four risk tiers. 2. Agentic AI only answers single prompts and cannot use external tools. 3. The guidelines call for a manual emergency stop for autonomous agents. Which are correct? **(Ans: 1 and 3 only)**

FQ – MAINS (Possible Question)

- "Agentic AI needs audits, not just ethics codes." Discuss with reference to CERT-In's guidelines. (250 words)

GS-III | Internal Security – Defence Technology (Missiles)

10. Pralay: India's Quasi-Ballistic Battlefield Missile Tested Again

Syllabus: *GS-III: Science and technology – indigenisation; security forces and their mandate.*

CONTEXT

DRDO flight-tested **Pralay**, an indigenous **surface-to-surface quasi-ballistic missile**, from **Dr APJ Abdul Kalam Island** off **Odisha**.



Illustration: a road-mobile canister launcher on a coastal range with a missile on a low arc over the sea and a radar dish behind.

KEYWORDS DEFINED

Keyword	Meaning
Quasi-ballistic	Flies a **low, manoeuvring path** instead of a high fixed arc – **harder to intercept**.
SRBM	**Short-range ballistic missile** – range up to about **1,000 km**.
Canisterised	Missile stored and fired from a **sealed tube** – **quick launch**, easy transport.
RLG-INS	**Ring Laser Gyro-based Inertial Navigation** – guides the missile without outside signals.
DcPP	**Development-cum-Production Partner** – industry builds the system alongside DRDO.

What Pralay is

- A **short-range, tactical** quasi-ballistic missile, fired **surface-to-surface** from a **sealed canister**, carrying **conventional warheads**.
- Built by **Research Centre Imarat (RCI), Hyderabad** with other DRDO labs, using technology from the **Prithvi Defence Vehicle** and **Prahaar**.
- **Bharat Dynamics Limited (BDL)** is the **production partner**.

Key features

- **Range 150–500 km**; warhead **500–1,000 kg**.
- **Single-stage solid-fuel** motor; **road-mobile** launcher for quick deployment.

- **Manoeuvring low trajectory** makes tracking and interception hard.
- **RLG-INS** plus **satellite navigation** for precision; high-explosive, fragmentation or penetration warheads.

Why it matters

- Fills the gap between **Pinaka rockets** and **longer-range missiles**.
- Strengthens **conventional deterrence** along the **LAC and LoC**.
- Purely **conventional** – a battlefield weapon, not a strategic nuclear one.

Test range

- **Dr APJ Abdul Kalam Island** (earlier **Wheeler Island**) lies off **Odisha's coast** in the **Bay of Bengal**.
- **Value addition:** India's **Integrated Guided Missile Development Programme (1983)** produced **Prithvi, Agni, Akash, Trishul and Nag** *(general knowledge)*.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **missile types, ranges, developers** and **test sites**.

PYQ – MAINS

- **Pattern:** Mains asks about **defence indigenisation** and **deterrence**.

FQ – PRELIMS (Possible Question)

- Pralay, recently flight-tested, is best described as: (a) a long-range nuclear ballistic missile (b) a short-range conventional quasi-ballistic missile (c) a surface-to-air missile (d) an anti-tank guided missile **(Ans: b)**

FQ – MAINS (Possible Question)

- "Conventional precision strike strengthens deterrence without nuclear escalation." Discuss with reference to Pralay. (150 words)

GS-III | Science & Technology – Space (Planetary Science)

11. Teghaza 001: A Pink Rock From Mars's Earliest Crust

Syllabus: *GS-III: Awareness in the field of space.*

> **CONTEXT**
>
> A **NASA-funded study** in the journal **Science** confirmed that **Teghaza 001**, an **800-gram rose-coloured meteorite** found in the **Sahara (Mali) in 2022**, came from **Mars** and formed **over 4.1 billion years ago**.



Illustration: a small pink meteorite on desert sand, with Mars and a streaking fireball in the sky and a magnifier showing tiny crystals.

KEYWORDS DEFINED

Keyword	Meaning
Meteorite	A rock from space that **survives its fall** to Earth's surface.
Noachian era	The **oldest period** of Martian history (about **4.1–3.7 billion years ago**), when **water** was common *(general knowledge)*.
Zircon	A tough mineral whose crystals **record a rock's age** through isotopes.
Rb–Sr dating	**Rubidium–strontium** dating – measures age from radioactive decay.
Oxygen isotopes	The **'fingerprint'** that tells which planet a rock came from.

What it is

- A rare **rose-coloured Martian meteorite** of about **800 g**.
- One of about **435 confirmed Martian meteorites** on Earth – called a **'Rosetta Stone' for early Mars**.
- Found in **Teghaza, Mali**, in the **Sahara** in **2022**.

Where it came from

- Crystallised from **cooling magma** in Mars's crust **over 4.1 billion years ago** – the **Noachian era**.
- Data matched with NASA's **Mars Reconnaissance Orbiter** point to **13 possible source areas**, mostly in Mars's **cratered southern highlands**.
- Thrown into space by a **huge impact**, it later crossed **Earth's orbit**.
- **Oxygen isotope** tests at the **University of New Mexico** confirmed its Martian origin.

What it tells us

- **Zircon crystals** keep an **isotope record** of its age.
- Scientists used **in-situ Rb–Sr dating** on tiny grains **without destroying** the rock.
- Signs that **liquid water** moved through Mars's crust about **4.1 billion years ago** – **hydrothermal activity**.

Why it matters

- Helps judge whether early Mars could have **supported microbial life**.
- A **Martian sample in the lab** without a costly **sample-return mission**.
- **Value addition:** NASA's **Perseverance** rover is collecting samples in **Jezero Crater**; India's **Mangalyaan (2013)** reached Mars orbit on its **first attempt** *(general knowledge)*.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **planetary missions, meteorites** and **dating methods**.

PYQ – MAINS

- **Pattern:** Mains rarely asks this directly; use it in **space science** answers.

FQ – PRELIMS (Possible Question)

- Teghaza 001, recently in news, is: (a) an exoplanet (b) a Martian meteorite found in the Sahara (c) a lunar sample from Chang'e-6 (d) a comet **(Ans: b)**

FQ – MAINS (Possible Question)

- "Meteorites are free sample-return missions." Discuss their value for planetary science. (150 words)

GS-III | Economy – Innovation and Indices

12. Global Innovation Index 2026: India at 38th, Top of Its Income Group

Syllabus: *GS-III: Indian economy; science and technology – developments and indigenisation.*

CONTEXT

WIPO released the **Global Innovation Index (GII) 2026**, ranking **139 economies**; **India is 38th**, leading **lower-middle-income** economies and the **Central and Southern Asia** region.



Illustration: a globe surrounded by rising bars, light bulbs, a patent seal, a rocket and gears, with podium blocks in front.

KEYWORDS DEFINED

Keyword	Meaning
WIPO	**World Intellectual Property Organization** – a **UN specialised agency** in **Geneva**.
GII	**Global Innovation Index** – ranks economies on **innovation inputs and outputs** (first published **2007**).
Innovation overperformer	A country whose innovation **beats what its income level predicts**.
S&T cluster	A **city-region** dense with **patents and scientific papers**.
Venture capital	**Equity funding** for young, high-growth firms.

How the index works

- Published by **WIPO** with the **Portulans Institute**; **19th edition** in 2026.
- **Seven pillars: input** – institutions, human capital and research, infrastructure, market sophistication, business sophistication; **output** – knowledge and technology outputs, creative outputs.
- Also maps the world's **top 100 S&T clusters**.

Global findings

- **Top 5: Switzerland, Sweden, United States, South Korea, Singapore**.
- Venture capital about **$510 billion in 2025**, with **AI taking over 53%**.
- World **R&D about $3.4 trillion** (about **2% of GDP**); **EVs 25%** of new car sales; **4.7 million** industrial robots.

- **Top cluster: Shenzhen–Hong Kong–Guangzhou**.

India's performance

- **38th overall**; **top lower-middle-income** economy and **top in Central and Southern Asia**.
- An **innovation overperformer** for **16 years in a row (2011–2026)**, with **Vietnam**.
- **15th** in **knowledge and technology outputs**, **1st** in **ICT services exports**, **4th** in **late-stage VC deals**.
- **Clusters: Bengaluru (25th), Delhi (27th), Mumbai (43rd), Chennai (79th)**.

Gaps and way forward

- Low **R&D spending** and weak **industry–academia links**.
- Push **ANRF** funding, **private R&D**, and **university–industry** partnerships.
- **Value addition:** India was **81st in 2015**; other income leaders – **China 10th** (upper-middle), **Rwanda 96th** (low) *(general knowledge for 2015)*.

PYQ – PRELIMS

- **Pattern:** Prelims asks **which body publishes which index** and India's rank.

PYQ – MAINS

- **Pattern:** Mains asks about India's **innovation ecosystem** and **R&D gaps**.

FQ – PRELIMS (Possible Question)

- The Global Innovation Index is published by: (a) the World Economic Forum (b) the World Bank (c) the World Intellectual Property Organization (d) UNCTAD **(Ans: c)**

FQ – MAINS (Possible Question)

- "India innovates above its income but below its potential." Examine using GII 2026. (150 words)

GS-III | Agriculture – Animal Husbandry; Disease Preparedness

13. Exercise VAJRA-SAT: Rehearsing for a Foot-and-Mouth Strain India Has Never Seen

Syllabus: *GS-III: Animal rearing economics; disaster management. GS-II: Health (One Health).*

CONTEXT

The **Department of Animal Husbandry and Dairying** held **Exercise VAJRA-SAT**, a **national tabletop exercise** to prepare for **foot-and-mouth disease (FMD) serotype SAT-1**, which has **never been reported in India** but has spread to parts of **West Asia**.



Illustration: a veterinary team in protective suits around a table map with cattle, buffalo, sheep and goat icons, a vaccine vial and a microscope.

KEYWORDS DEFINED

Keyword	Meaning
Tabletop exercise	A **discussion-based drill** on a **simulated outbreak** – no field action.
Serotype	A **strain variant** of a virus; vaccines must **match** it.
FMD	**Foot-and-mouth disease** – a highly contagious **viral** disease of **cloven-hoofed** animals.
WOAH	**World Organisation for Animal Health** (founded **1924** as **OIE**).
Transboundary disease	An animal disease that **spreads across borders** quickly.

About the exercise

- Organised by **DAHD** with **ICAR-NIVEDI, Bengaluru**.
- Theme: **'Detect Early, Confirm Fast, Contain Together'**.
- **83 participants** from **25 States and 3 UTs**, plus labs, universities, vaccine makers, and **defence and border agencies**.
- Practised **early detection, lab confirmation, movement control, emergency vaccination** and **risk communication**.

About FMD

- Affects **cattle, buffaloes, sheep, goats and pigs**; caused by an **Aphthovirus**.
- Causes **blisters**, lower **milk yield**, slow growth, **infertility** and weak **draught power**.
- Spreads via **animals, secretions, air, meat, milk products, equipment and clothes**.
- **Not a human health threat** – different from **hand, foot and mouth disease** of children.

Why it matters

- FMD triggers **trade bans** on livestock products – a big blow to **dairy and meat exports**.
- First disease for which **WOAH** gave **official status recognition**.
- A **new serotype** would make existing vaccines **less useful** – hence **preparedness**.

India's FMD control

- **Mass vaccination** at regular intervals until cases fall.
- **Value addition:** the **National Animal Disease Control Programme (2019)** aims to control **FMD and brucellosis** with **100% vaccination** of cattle, buffalo, sheep, goats and pigs; India has the **world's largest livestock population** *(general knowledge)*.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **livestock diseases**, **vaccination programmes** and **animal-health bodies**.

PYQ – MAINS

- **Pattern:** Mains asks about **livestock economy** and **transboundary disease** preparedness.

FQ – PRELIMS (Possible Question)

- Exercise VAJRA-SAT was conducted to prepare against: (a) avian influenza H5N1 (b) African swine fever (c) foot-and-mouth disease serotype SAT-1 (d) lumpy skin disease **(Ans: c)**

FQ – MAINS (Possible Question)

- "Preparedness is cheaper than an outbreak." Discuss with reference to transboundary animal diseases. (150 words)

GS-III | Security – Military Exercises; India–Myanmar Relations

14. RAHAT MITRA 2026: India and Myanmar's First Tri-Services Disaster Drill

Syllabus: *GS-III: Disaster management; security forces. GS-II: India and its neighbourhood.*

CONTEXT

RAHAT MITRA 2026, held in **Chennai**, was the **first HADR (humanitarian assistance and disaster relief) exercise** in which all **three services** of **India and Myanmar** trained together.



Illustration: a flood relief operation with army engineers laying a small bridge, a navy ship offshore, a helicopter dropping supplies and rescue boats.

KEYWORDS DEFINED

Keyword	Meaning
HADR	**Humanitarian Assistance and Disaster Relief** – military help in disasters.
Tri-services	Involving the **Army, Navy and Air Force** together.
Interoperability	The ability of forces to **work together smoothly**.
Act East Policy	India's policy (**2014**) of deeper ties with **South-East Asia and the Indo-Pacific**.
Engineer Task Force	Army units that **build bridges, clear debris** and restore access.

About the exercise

- **First** bilateral **tri-services HADR** exercise between **India and Myanmar**.
- Held over **two days** in **Chennai**; co-hosted by both and coordinated by the **Indian Navy**.
- Theme: **'Together for Humanity, Stronger through Partnership'**.
- **Over 300 personnel** from both sides' **Army, Navy and Air Force**.

What was practised

- Professional exchanges and **interoperability** in disaster response.
- The Indian Army showed its **Flood Relief Columns** and **Engineer Task Forces**.
- Demonstrations of **rescue, dewatering** and **bridge-laying** to restore access.

Other India–Myanmar exercises

- **IMBEX** – India–Myanmar Bilateral Military Exercise (Army).
- **IMNEX** – India–Myanmar Naval Exercise.

Why it matters

- Both lie in **cyclone and earthquake-prone** regions – the Bay of Bengal links them.
- Builds trust with a **strategically placed neighbour** – **Act East** and **neighbourhood first**.
- **Value addition:** India sent relief to Myanmar under **Operation Brahma** after the **March 2025 earthquake**; India and Myanmar share a **1,643 km** border across **Arunachal, Nagaland, Manipur and Mizoram** *(general knowledge)*.

PYQ – PRELIMS

- **Pattern:** Prelims pairs **exercises with partner countries**.

PYQ – MAINS

- **Pattern:** Mains asks about **India–Myanmar ties** and **HADR diplomacy**.

FQ – PRELIMS (Possible Question)

- Consider the following pairs (Exercise : Partner): 1. RAHAT MITRA – Myanmar 2. IMNEX – Myanmar 3. VAJRA-SAT – Sri Lanka. How many are correctly matched? **(Ans: Only two)**

FQ – MAINS (Possible Question)

- "HADR is India's soft-power bridge in the neighbourhood." Discuss. (150 words)

GS-III | Internal Security – Defence Indigenisation (Navy) UPDATE

15. Update: Amini Joins the Arnala Class of Submarine Hunters

Syllabus: *GS-III: Security challenges; indigenisation of technology.*

WHAT WE KNOW TILL NOW

- **GRSE, Kolkata** delivered **'Amini'**, an **Anti-Submarine Warfare Shallow Water Craft**, to the Navy; **over 80% indigenous**.
- Named after **Amini Island, Lakshadweep**; the old **INS Amini** (a **Petya III** corvette) was retired in **2002**.

WHAT TODAY ADDS

- It belongs to the **Arnala class** and will **replace the ageing Abhay-class corvettes**.
- Fitted with **hull-mounted sonar**, a **Low-Frequency Variable Depth Sonar** and a **30 mm naval gun**; built to **Indian Register of Shipping** standards.

CONTEXT

New details show **Amini** belongs to the **Arnala class**, built to replace the **Abhay-class corvettes**, with advanced **sonar** for hunting submarines in **shallow coastal waters**.



Illustration: a small grey warship on waterjets near a palm-fringed island, with sonar waves reaching a submarine outline below.

KEYWORDS DEFINED

Keyword	Meaning
Arnala class	India's new class of **ASW Shallow Water Craft**; **INS Arnala** was the first (**2025**) *(general knowledge)*.
Abhay class	Older **anti-submarine corvettes** now being **replaced**.
LFVDS	**Low-Frequency Variable Depth Sonar** – lowered **below the hull** to hear submarines in tricky water layers.
Hull-mounted sonar	Sonar **fixed to the ship's hull**.
IRS	**Indian Register of Shipping** – sets **ship construction standards**.

What is new

- Part of the **Arnala class** (GRSE series) of ASW shallow water craft.
- Will **replace** the ageing **Abhay-class corvettes**.
- Built to **Indian Register of Shipping (IRS)** classification rules.

Sensors and weapons

- **Hull-mounted sonar** plus **Low-Frequency Variable Depth Sonar (LFVDS)** for hard shallow-water conditions.
- **Anti-submarine rockets** and **lightweight torpedoes**, including indigenous systems.
- A **30 mm naval gun**; indigenous **combat management system**, sensors and navigation.

Design

- **Waterjet propulsion** – speed, agility and **less underwater noise**.
- Missions: **hunt submarines**, **coastal surveillance**, **defensive mine-laying**.

Significance

- Protects **naval bases, harbours** and coasts from **submarines and underwater drones**.
- Shows India's growing **warship-building** capacity – **Aatmanirbhar Bharat**.
- **Value addition:** India plans **16** such craft – **8 by GRSE** and **8 by Cochin Shipyard** *(general knowledge)*.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **ship classes, builders and sonar** types.

PYQ – MAINS

- **Pattern:** Mains asks about **littoral security** and **defence indigenisation**.

FQ – PRELIMS (Possible Question)

- The Arnala-class ASW Shallow Water Craft are meant to replace which older ships? (a) Kamorta-class corvettes (b) Abhay-class corvettes (c) Shivalik-class frigates (d) Kolkata-class destroyers **(Ans: b)**

FQ – MAINS (Possible Question)

- "Shallow waters are where submarines hide best." Discuss India's response. (150 words)

GS-III | Science & Technology – Nuclear Fusion UPDATE

16. Update: Tokamak – Origins, Heating and Safety

Syllabus: *GS-III: Science and technology; energy.*

WHAT WE KNOW TILL NOW

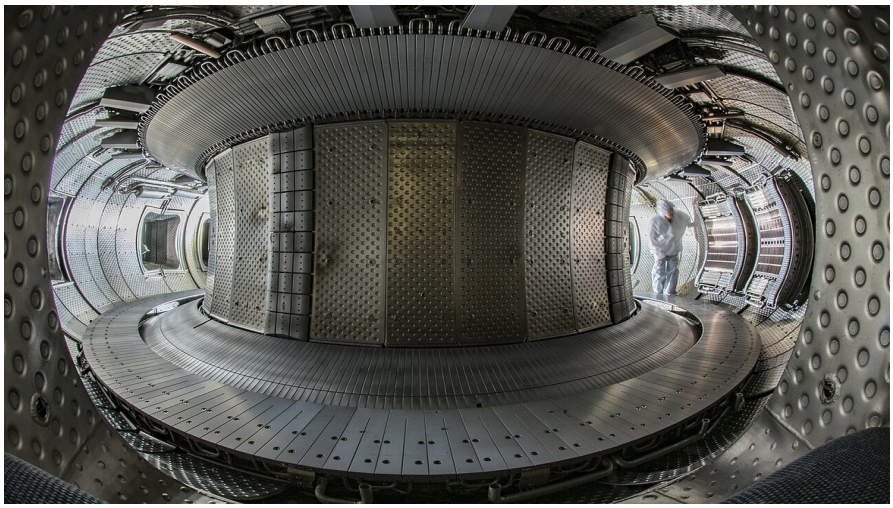
- A **tokamak** holds super-hot **plasma** in a **ring-shaped chamber** using a **twisted (helical) magnetic field**.
- **ITER (France)**, **China's EAST (1,066 seconds, 2025)**, India's **ADITYA-U, SST-1** and the private **PRAGYA**; **no tokamak** has made **net electricity**.

WHAT TODAY ADDS

- **Origins:** a **Russian acronym** for 'toroidal chamber with magnetic coils', conceived in the **early 1950s** by **Igor Tamm and Andrei Sakharov**.
- **Heating** above **100 million °C**, **energy density** and why fusion is **inherently safe**.

CONTEXT

As the race for **fusion power** continues, more is being explained about **how tokamaks heat their fuel**, why they are **safe**, and where the design came from.



Inside the WEST tokamak in France: the ring-shaped steel vacuum chamber where plasma is held by magnetic fields.
Photo: Christophe Roux / IRFM, Wikimedia Commons, CC BY 4.0

KEYWORDS DEFINED

Keyword	Meaning
Tokamak (origin)	Russian acronym – **'toroidal chamber with magnetic coils'**.
Ohmic heating	Heating plasma by **driving an electric current** through it.
Neutral beam injection	Firing **high-energy neutral atoms** into plasma to heat it.
Poloidal field	The magnetic field that runs **the short way around** the ring; with the **toroidal** field it makes a **helix**.
Runaway reaction	An **uncontrolled chain reaction** – possible in fission, **not in fusion**.

Where it came from

- A **Russian acronym** meaning **toroidal chamber with magnetic coils**.
- Proposed by **Andrei Sakharov and Igor Tamm**, physicists of the **Soviet Union**, in the **early 1950s**.

How fuel is heated

- **Deuterium–tritium** fuel is heated **above 100 million °C**.
- Three methods: **electric current (ohmic)**, **radio-frequency waves**, and **high-energy neutral beams**.
- **Toroidal + poloidal** fields make the **helical** path that limits drift.

Energy and fuel

- D–T fusion releases about **3.4×10^{14} joules per kilogram** – vastly more than fossil fuels.
- A **lithium blanket** absorbs neutrons, makes **heat for turbines** and **breeds tritium**.

Safety and limits

- **Inherently safe:** only a **tiny amount of fuel** is in the chamber; any fault **stops fusion** – **no runaway chain reaction**.
- **Limits:** turbulence and **instabilities**, **neutron damage** to walls, costly **cooling and magnets**.
- **Value addition: fission** splits heavy atoms like **uranium-235**; fusion joins light ones like **hydrogen isotopes** *(general knowledge)*.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **fusion vs fission**, **ITER** and **plasma physics** terms.

PYQ – MAINS

- **Pattern:** Mains asks about **clean energy** and India's **fusion research**.

FQ – PRELIMS (Possible Question)

- The tokamak concept was first proposed in the 1950s by scientists of: (a) the United States (b) the Soviet Union (c) France (d) Japan **(Ans: b)**

FQ – MAINS (Possible Question)

- "Fusion promises clean power without meltdown risk." Examine. (150 words)

GENERAL STUDIES PAPER IV

[GS-IV] Ethics, Integrity & Aptitude

GS-IV | Ethics, Integrity & Aptitude – Synthesis

17. Ethics Synthesis: Liberty, Care and Responsible Machines

Syllabus: *GS-IV: Ethics in public administration; accountability; ethical issues in technology; case studies.*

CONTEXT

Today's topics raise three values for public servants: **respect for liberty** (the 24-hour rule), **care for the vulnerable** (TB patients, affordable surgery), and **responsibility for technology** (AI agents and pollution data).



Illustration: a balance with a clock and gavel on one side, and a lung symbol and robot shield on the other.

KEYWORDS DEFINED

Keyword	Meaning
Due process	Fair **legal procedure** before taking away **liberty or property**.
Compassion	**Feeling for others' suffering** and acting to reduce it.
Accountability	Being **answerable** for decisions and their results.
Precautionary principle	Act to **prevent serious harm** even when science is **not yet certain**.
Evidence-based policy	Decisions grounded in **accurate data**, not convenience.

The ethical thread

- **Liberty:** the Court refused to let a **label ('not yet arrested')** hide real detention – **substance over form**.
- **Care:** REACH and Dr Shetty show **compassion turned into systems** – community TB care, low-cost surgery.
- **Responsibility:** AI agents need **kill switches and human approval**; pollution policy needs **honest data**, even when it is **inconvenient**.

Values at stake

- **Due process, compassion, accountability, transparency, integrity, precaution, objectivity, empathy.**

Case study 1: The late arrest entry

- **Situation:** As **SP**, you find your team held a suspect **overnight** and recorded the arrest **next morning** to 'gain time'.
- **Issues: liberty vs investigation**; **honesty in records**.
- **Options:** (a) let it pass; (b) quietly fix the time; (c) **record the true detention time**, produce the suspect **within 24 hours of it**, and **train the team**.
- **Preferred course: (c)** – **legality and honesty**; shortcuts **taint evidence**.

Case study 2: The AI agent rollout

- **Situation:** As head of a **government IT project**, you are pressed to launch an **AI agent** that can **approve refunds** before audit.
- **Issues: speed vs safety**; **accountability** for wrong payouts.
- **Options:** (a) launch now; (b) drop it; (c) **launch in a sandbox** with **human approval** for payouts, a **kill switch** and an **independent audit**.
- **Preferred course: (c)** – innovate **responsibly**.

Case study 3: The pollution figures

- **Situation:** As a **pollution officer**, new satellite data show **far more crop fires** than official counts – which may **embarrass** your department.
- **Issues: truth vs reputation**; **public health**.
- **Options:** (a) ignore the data; (b) leak it; (c) **report it formally**, explain the gap, and **update the inventory**.
- **Preferred course: (c)** – **integrity and evidence-based policy**.

Thinkers to quote (paraphrased)

- **Justice V.R. Krishna Iyer:** personal liberty is the **heart of the Constitution**.
- **Immanuel Kant:** treat people **as ends**, never merely as **means**.
- **Mahatma Gandhi:** recall the **poorest face** before acting.
- **Hans Jonas:** with **powerful technology** comes a duty of **foresight**.
- **Swami Vivekananda: service to people** is service to God.

PYQ – PRELIMS

- **Not applicable** – GS-IV is Mains-only.

PYQ – MAINS

- **Pattern:** GS-IV asks about **integrity in records**, **ethics of AI** and **compassion in public service**.

FQ – PRELIMS (Possible Question)

- **Not applicable** – GS-IV is Mains-only.

FQ – MAINS (Possible Question)

- "Substance, not labels, must decide when rights apply." Discuss in the context of police custody. (150 words)
- Case study: As head of an IT project pressed to launch an unaudited AI agent, what would you do and why? (250 words)